

Gate C: complete governed source processing path

Progress report, system-test record and show-and-tell

Evidence field	Recorded value
Status	Final - subject to review sign-off by exception
Delivery scope	Gate C completion candidate for DS-03 steps 7 and 9; KNO-01 processing and the complete acceptance path
Walkthrough	3 September 2026, retained local synthetic Minikube environment
Candidate source revision	1227b2c847d1cfc3d9db8860497703b9f5f4d29e+working-tree
New runtime image	public-purpose-lab/m3-runtime:development; sha256:39e6b046326dbdce266c5293c3492ff60867154b2cd5ef86f7a6650ef3d1fb37
Reused A-002 evidence image	sha256:37bbdbbe1a397270d97f2b9f1352ade3e4335b1915be589a6d98deefc671be79; unaffected validation and staging claims only
Scenario package	1.3.0; package sha256:7560c52f2aa49842831e66471d0bd2af81410d34b52f2e232dd0ae911fde416b; scenario sha256:a8c7df534f40030f372374ca0f8fb46ad7c86308df2aacad1ff5befa9fee0ec8
Environment, profile and trust	environment-c5ef1307484826d1b598ad73a08d31b6; public-purpose-lab-gate-a; environment-local synthetic root
Evidence session	session:eeb76d6f-4cec-4d8c-9393-c403a8bc92e9
Source and processing record	source-version:3e1f57e3-03b8-467f-a558-aed00df42b3f; processing:21169089-f0f6-49bb-9ba8-b311959de286
Restarted KNO-01 pod	gate-a-knowledge-processing-94845588f-2fh56; UID 419edab4-08a2-4482-b702-3bc1bdfc2211
Clean successor	session:dd40a28a-3182-4080-9bd2-8abe809a4e67; preparing
Publication evidence	Implementation, report and PDF are published together; the exact commit and hosted run are recorded in the publication handoff

This is synthetic, in-development demonstrator evidence. It does not establish production, legal, regulatory, clinical, government, NHS, charity, evidence-quality or compliance assurance, and it does not transfer accountable human authority to the Lab.

Outcome

The smallest complete Gate C finishing slice is implemented and demonstrated. KNO-01 consumes the durable source . staged fact for one exact immutable source version, records receipt idempotently in its own persistent store and emits processing . accepted, processing . started and one conclusive processing . completed or processing . failed outcome. Processing is deliberately elementary: it verifies the digest, counts bytes, lines and Markdown-like sections, and creates a bounded safe preview for the authorised Workbench.

The source body crosses only the protected CNT -01 to KNO-01 processing input boundary. It is absent from lifecycle events, general logs, Operations and Presentation. Presentation receives read-only progress, structural counts and explicit limitations; it has no command that can submit, validate, stage or process a source. The named reviewer remains responsible for the release request and no processing result is treated as approval of evidence or a finding.

The walkthrough reused the already accepted A-002 validation and reviewer-controlled staging behaviour. Its evidence is recorded in gate-c-validation-and-staging-show-and-tell .md and was not recreated. The new image and the evidence below cover only the additional KNO-01 runtime, projections, restart behaviour and complete DS-03 acceptance path.

Acceptance summary

Gate C finishing requirement	Evidence	Position
Consume the exact immutable staged-source fact	K-001 contract, component tests, Figures 2 and 4	Achieved
Record idempotent receipt and visible accepted, started and terminal states	Persistent KNO-01 store, Figures 2 and 4	Achieved
Keep processing basic and inspectable	Digest verification, 156 bytes, 7 lines, 2 sections and bounded preview in Figure 2	Achieved
Keep source bodies inside the authorised component boundary	Event, log and Presentation absence assertions; Figures 3 and 4	Achieved
Show processing state and bounded result in Workbench	Figure 2	Achieved
Show meaningful read-only progress in Presentation	Figure 3	Achieved
Show correlated CNT-01 to KNO-01 timings and safe evidence references	Figure 4	Achieved
Reconcile after KNO-01 restart without a second terminal result	Rollout restart, persistent processing identity and terminal count in Figure 5	Achieved
Prevent refused or unvalidated sources reaching KNO-01	Contract and end-to-end adverse cases	Achieved
Preserve component-owned semantics for upload and paste	End-to-end paste and upload paths	Achieved

Walkthrough flow

1. Director starts the admitted Gate C path

The Director evaluated the retained synthetic environment, created the Demonstration Session, prepared the admitted scenario, established the synthetic reviewer and audience bindings, and started DS-03. It then requested the target-owned Workbench source-intake view. The Director coordinates the story; it does not perform or overwrite CNT-01 or KNO-01 business operations.

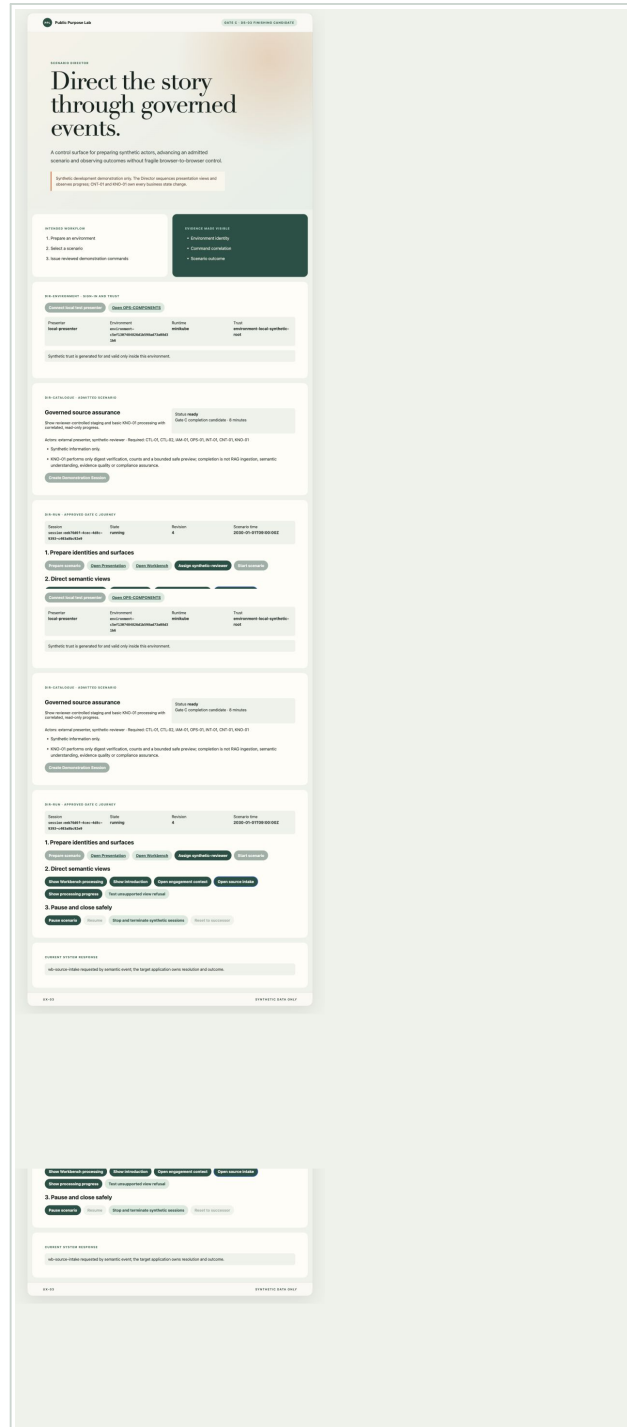


Figure 1. Running Gate C Demonstration Session with the retained environment identity, trust boundary and component-owned controls visible.

2. Workbench shows the bounded processing result

The reviewer used paste intake for a synthetic Markdown fixture and released the exact validated version through the existing A-002 path. Workbench then showed the KNO-01 record and its accepted, processing and completed stages. The single result verified digest

77db273194d2999b13a03a647da8152a2d0440d87c7ead6307ce5f980d5b1fcd, counted 156 bytes, 7 lines and 2 sections, and returned a bounded preview only to this authorised reviewer surface.

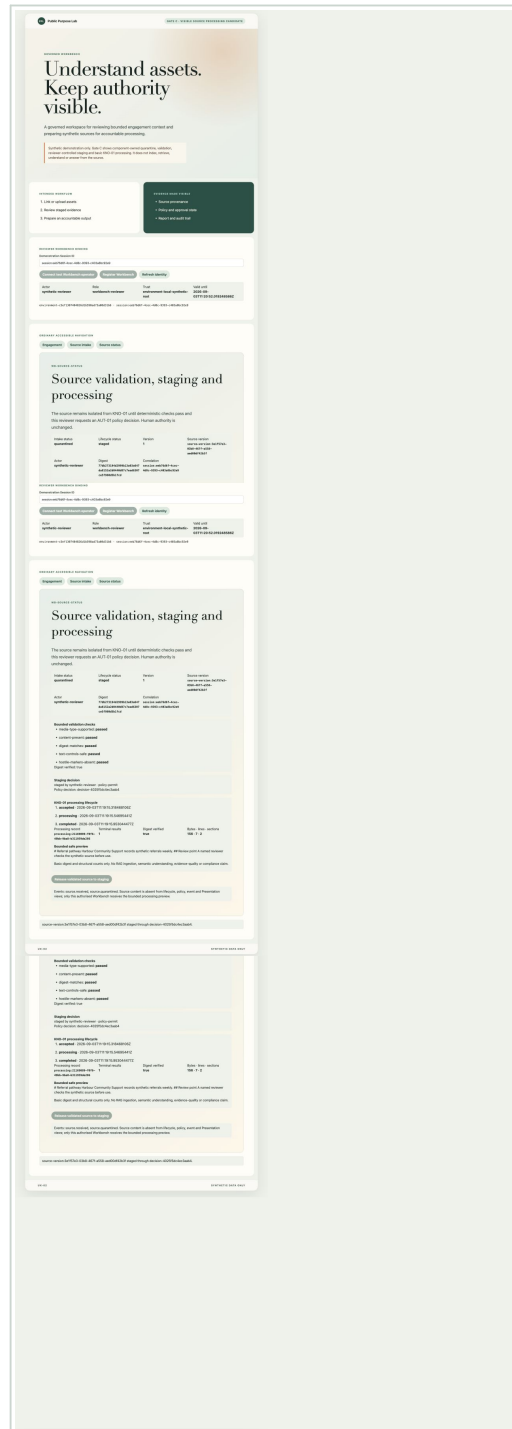


Figure 2. Workbench showing the exact staged source, KNO-01 lifecycle, one terminal result, structural counts and bounded safe preview.

3. Presentation reports progress without control or content

The Director requested PRES-PROGRESS. Presentation resolved the target-owned view from the component state and displayed KNO-01, completed and the three structural counts. It exposed neither the source body nor the preview and offered no business-operation control. The limitation explicitly rules out RAG ingestion, semantic understanding, evidence quality and compliance.

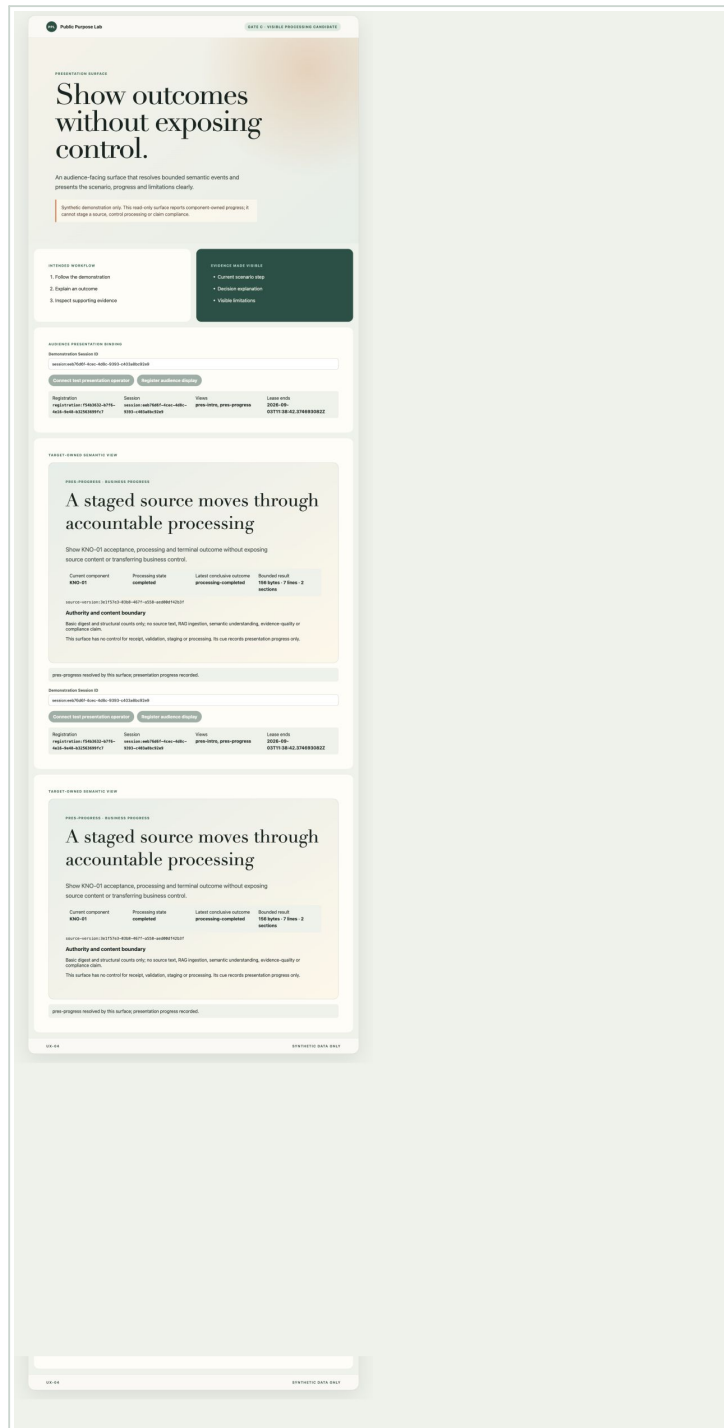


Figure 3. Presentation showing read-only KNO-01 progress, counts and limitations without source content or business controls.

4. Operations shows the correlated component sequence

Operations projected the complete sequence under correlation

session:eeb76d6f-4cec-4d8c-9393-c403a8bc92e9: CNT-01 receipt, quarantine, validation and staging, followed by KNO-01 acceptance, start and completion. Each row includes timestamp, component, event type, safe event reference and exact source-version subject. No event contains source content. Refused paths use the same safe projection and are covered by the adverse system tests.

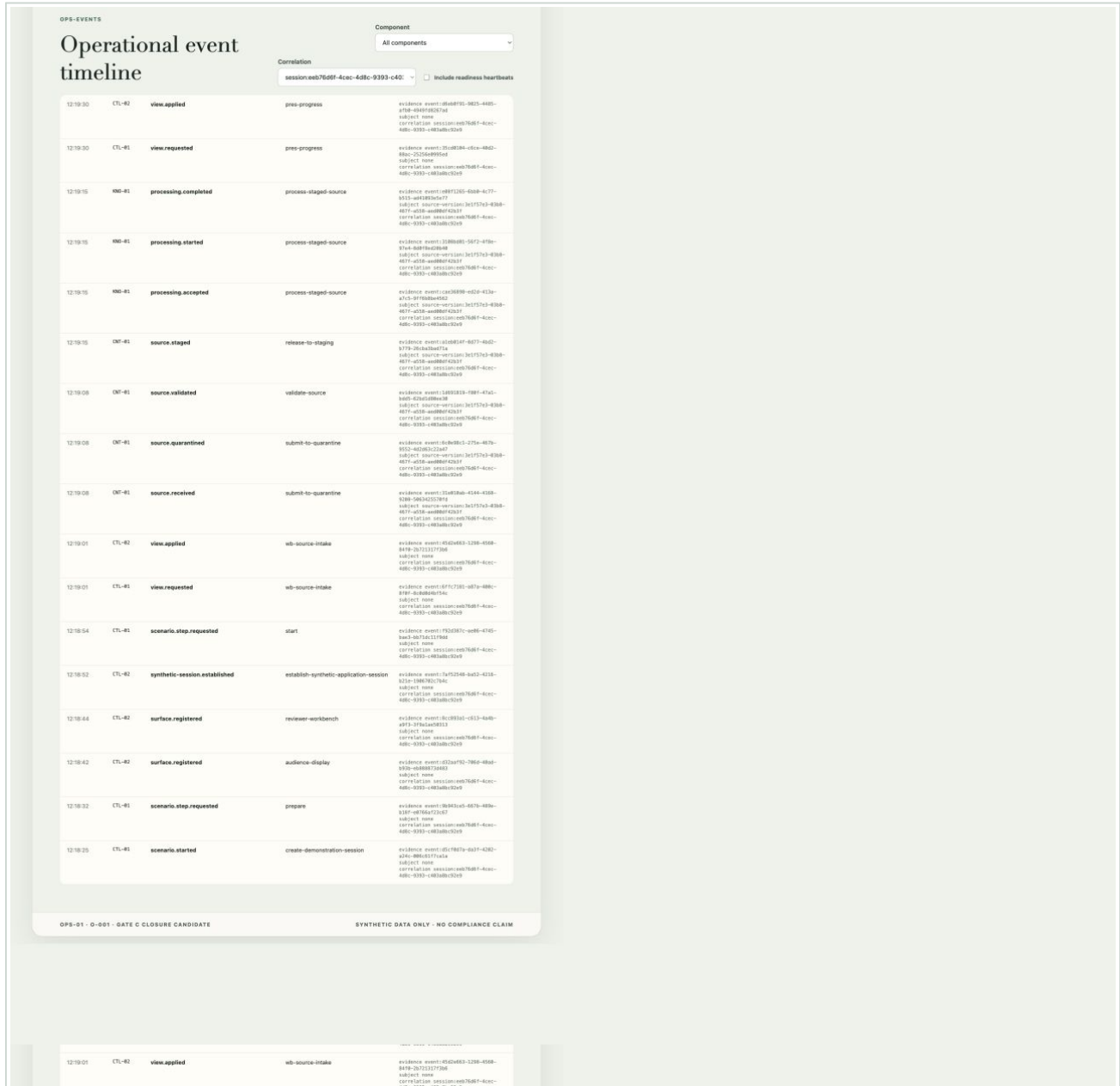


Figure 4. Correlated Operations timeline showing CNT-01 source.staged followed by KNO-01 processing.accepted, processing.started and processing.completed.

5. Restart reconciliation preserves one conclusive result

The gate-a-knowledge-processing deployment was restarted after completion. Kubernetes replaced the workload and reported the new pod ready with the same image digest. KNO-01 reopened its retained SQLite state, reconciled the staged work and returned the same processing identity, original stage timestamps and Terminal results: 1. It did not emit or record a second completion.

```
deployment.apps/gate-a-knowledge-processing restarted
deployment "gate-a-knowledge-processing" successfully rolled out
new pod UID: 419edab4-08a2-4482-b702-3bc1bdfc2211
processing record: processing:21169089-f0f6-49bb-9ba8-b311959de286
terminal results after restart: 1
```

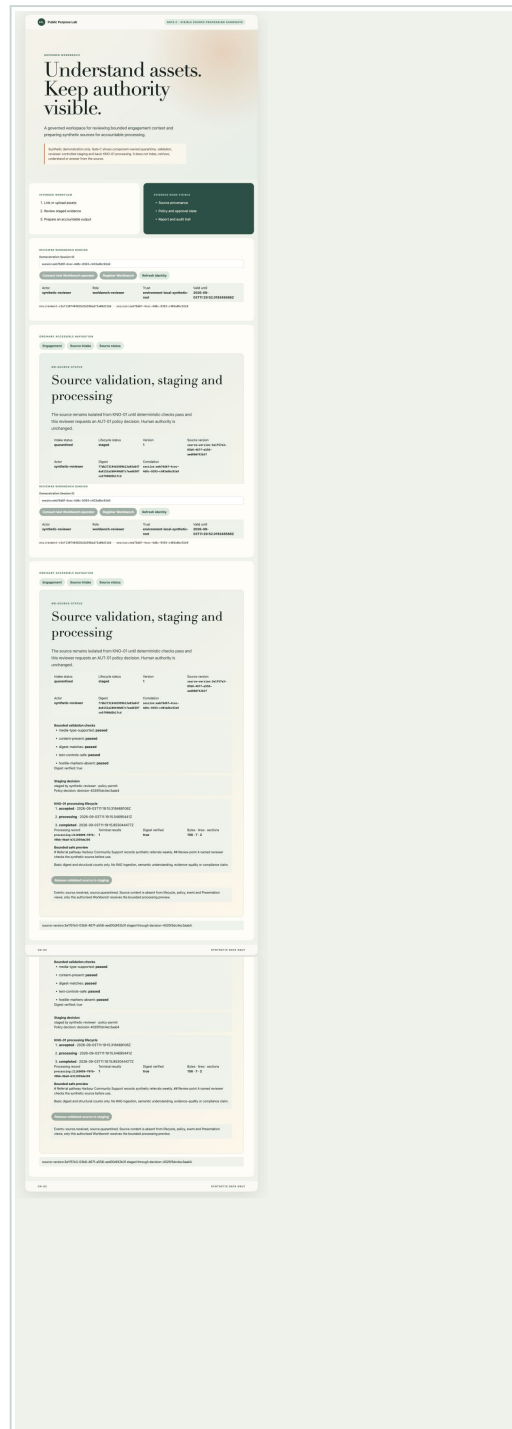


Figure 5. Workbench after KNO-01 rollout restart, preserving the same processing record, timestamps, bounded result and single terminal count.

Adverse, contract and component evidence

Evidence layer	Conclusive checks
K-001 contract	Versioned content query, protected response, lifecycle fact and status projections; duplicate-terminal fixture rejected
KNO-01 component	Exact-source idempotency, reopen reconciliation, one terminal result, bounded completed result and safe disclosed failure
CNT-01 component	Only the exact staged version is returned to the environment-bound KNO-01 purpose; wrong session, caller or lifecycle is refused
End-to-end paste	Quarantine, validation, staged fact, accepted, started, completed, bounded result and one terminal outcome
End-to-end upload	The same component-owned lifecycle and terminal semantics as paste
Refusals	Empty, malformed, NUL-containing, oversized and hostile fixtures do not reach KNO-01; an unvalidated staging attempt is refused and has no processing record
Failure	A disclosed synthetic processing-failure fixture reaches one safe processing . failed result without body disclosure
Privacy	Exact fixture body absent from Operations events, Presentation responses and CNT-01/KNO-01 general logs
Restart	Real Minikube rollout restart returns the same processing identity with terminal count one

The retained Minikube node exhausted its image filesystem during the candidate rebuild. Unused build images were pruned with the container runtime's standard image-prune operation; persistent volumes, the environment trust material and retained business state were not removed. The affected SQLite-owning workloads then recovered. This is an operator-capacity lesson, not a source-processing failure or a reason to weaken fail-closed persistence behaviour.

Functions, rules and contracts delivered

Area	Current Gate C baseline
KNO - 01	Component-owned persistent receipt, lifecycle, bounded processor, transactional outbox and startup reconciliation
K - 001	Versioned protected-content query and metadata-only lifecycle/status contracts with examples and negative fixtures
CNT - 01	Exact staged-body response restricted to the authorised KNO-01 environment, session and processing purpose
Workbench UX - 02	Lifecycle stages, exact processing record, one terminal count, digest/count result, bounded preview and limitations
Presentation UX - 04	Read-only component progress and structural counts with no source text, preview or business controls
Operations OPS - 01	Correlated CNT-01 to KNO-01 lifecycle, timings, refusals and safe event/subject references
INT - 01 and deployment	Durable Gate C subjects, KNO-01 identity permissions and a Kubernetes persistent processing-state volume
Assurance	Paste/upload parity, adverse-input exclusion, failure path, content-absence assertions and real restart reconciliation

No new ADR was required. The slice uses the existing component mesh, versioned event envelope, environment-scoped workload identity, NATS JetStream, component-owned SQLite persistence and Kubernetes

deployment conventions. It adds a contract and persistent volume only where the demonstrated KNO-01 path needs them; it does not create a new platform or technology commitment.

Gate position

The implementation and local evidence satisfy the complete DS-03 Gate C acceptance path. Publication and a successful exact-commit hosted quality gate complete the engineering evidence. Formal Gate C closure remains a founder governance action under review by exception; this report does not pre-empt that sign-off.